

## POINT LEPREAU, NB, Canada

WMO: 716990

Lat: 45.059N

Lon: 66.459W

Elev: 20

StdP: 14.69

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -1.0       | 3.5 | -12.2                         | 2.8 | 0.9  | -7.8 | 3.6 | 6.0  | 31.8                  | 40.9 | 27.7 | 38.5 | 10.8                  | 320  | 0.605 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.9                   | 74.0            | 63.6 | 70.8 | 62.5 | 68.2 | 61.6 | 66.1                | 70.4 | 64.3 | 68.3 | 62.9 | 66.4 | 7.5                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|----------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)            |
| 64.5                            | 90.8 | 68.1 | 62.8 | 85.5 | 66.0 | 61.4 | 81.3 | 64.4 | 30.9          | 70.6 | 29.5 | 68.5 | 28.5 | 66.6 | 73.4           |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 23.1          | 19.4 | 16.6 | DB                         | -6.6 | 82.0               | 4.5 | 2.9                                                | -9.8  | 84.1       | -12.5 | 85.7       | -15.0 | 87.4       | -18.3 | 89.5 | (4) |
| (5)               |      |      | WB                         | -7.8 | 70.8               | 4.3 | 1.6                                                | -10.9 | 72.0       | -13.4 | 72.9       | -15.8 | 73.8       | -19.0 | 75.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 43.5   | 22.9  | 24.7  | 31.5  | 40.1  | 48.3  | 55.6  | 60.5  | 61.7  | 57.3  | 49.0  | 39.8  | 29.8  |
|                                                                               | DBStd       | 14.97  | 11.02 | 9.30  | 7.57  | 4.90  | 4.45  | 4.49  | 3.65  | 3.23  | 4.65  | 5.90  | 7.69  | 9.37  |
|                                                                               | HDD50       | 3544   | 840   | 709   | 573   | 297   | 86    | 5     | 0     | 0     | 4     | 92    | 312   | 626   |
|                                                                               | HDD65       | 7857   | 1305  | 1129  | 1037  | 746   | 518   | 285   | 149   | 112   | 234   | 496   | 755   | 1091  |
|                                                                               | CDD50       | 1183   | 0     | 0     | 0     | 1     | 33    | 172   | 324   | 362   | 223   | 60    | 8     | 0     |
|                                                                               | CDD65       | 22     | 0     | 0     | 0     | 0     | 0     | 2     | 8     | 9     | 3     | 0     | 0     | 0     |
|                                                                               | CDH74       | 97     | 0     | 0     | 0     | 0     | 4     | 14    | 31    | 35    | 12    | 1     | 0     | 0     |
|                                                                               | CDH80       | 7      | 0     | 0     | 0     | 0     | 0     | 1     | 3     | 2     | 1     | 0     | 0     | 0     |
| Wind                                                                          | WSAvg       | 7.5    | 9.6   | 9.2   | 8.9   | 7.8   | 6.6   | 5.7   | 4.8   | 5.0   | 6.1   | 8.0   | 9.0   | 9.7   |
| Precipitation                                                                 | PrecAvg     | 49.7   | 4.0   | 3.6   | 3.8   | 3.9   | 4.1   | 4.4   | 3.8   | 3.5   | 4.0   | 5.0   | 4.8   | 5.0   |
|                                                                               | PrecMax     | 65.7   | 7.1   | 7.4   | 7.6   | 7.7   | 8.4   | 10.6  | 9.0   | 7.9   | 9.1   | 12.2  | 8.3   | 10.2  |
|                                                                               | PrecMin     | 36.3   | 1.9   | 1.3   | 0.6   | 0.9   | 1.9   | 1.6   | 1.2   | 0.7   | 1.0   | 1.4   | 2.2   | 2.1   |
|                                                                               | PrecStd     | 7.4    | 1.3   | 1.2   | 1.8   | 1.7   | 1.6   | 2.3   | 2.0   | 1.7   | 2.2   | 2.5   | 1.6   | 1.9   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 49.8  | 45.9  | 52.3  | 59.0  | 69.5  | 75.9  | 78.0  | 78.0  | 74.8  | 65.9  | 58.1  | 52.8  |
|                                                                               |             | MCWB   | 48.9  | 43.6  | 42.5  | 46.2  | 54.9  | 61.0  | 66.3  | 64.9  | 64.8  | 58.2  | 55.4  | 51.4  |
|                                                                               | 2%          | DB     | 46.6  | 42.7  | 46.4  | 53.0  | 61.8  | 70.5  | 73.1  | 73.8  | 69.4  | 61.4  | 55.9  | 50.3  |
|                                                                               |             | MCWB   | 45.2  | 40.9  | 41.5  | 44.3  | 52.8  | 60.1  | 64.2  | 64.2  | 61.2  | 57.4  | 53.9  | 48.9  |
|                                                                               | 5%          | DB     | 43.4  | 40.7  | 43.6  | 50.0  | 58.2  | 66.3  | 69.7  | 70.6  | 66.5  | 59.4  | 54.0  | 47.5  |
|                                                                               |             | MCWB   | 42.0  | 38.9  | 40.2  | 43.5  | 51.4  | 58.5  | 63.1  | 63.2  | 60.6  | 56.1  | 52.1  | 45.9  |
|                                                                               | 10%         | DB     | 39.7  | 38.5  | 41.2  | 47.4  | 55.2  | 63.0  | 66.8  | 68.0  | 64.2  | 57.8  | 52.0  | 44.4  |
|                                                                               |             | MCWB   | 37.8  | 36.5  | 38.5  | 42.5  | 49.8  | 57.3  | 61.7  | 62.3  | 59.8  | 54.8  | 49.8  | 42.1  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 49.1  | 44.7  | 46.6  | 50.5  | 59.4  | 65.3  | 69.9  | 69.0  | 66.5  | 61.1  | 56.8  | 51.9  |
|                                                                               |             | MCDB   | 49.6  | 45.4  | 48.1  | 54.0  | 64.3  | 71.3  | 74.9  | 73.6  | 70.1  | 63.3  | 57.3  | 52.5  |
|                                                                               | 2%          | WB     | 45.4  | 41.4  | 43.1  | 47.0  | 55.3  | 62.1  | 66.2  | 66.1  | 63.9  | 58.9  | 54.6  | 49.4  |
|                                                                               |             | MCDB   | 46.3  | 42.2  | 44.8  | 49.9  | 59.7  | 67.2  | 70.4  | 70.5  | 66.8  | 60.1  | 55.4  | 50.1  |
|                                                                               | 5%          | WB     | 42.2  | 39.2  | 41.0  | 45.1  | 52.8  | 59.8  | 63.9  | 64.4  | 62.4  | 57.3  | 52.5  | 46.2  |
|                                                                               |             | MCDB   | 43.2  | 40.2  | 42.8  | 48.0  | 56.4  | 64.1  | 68.0  | 68.4  | 64.9  | 58.7  | 53.6  | 47.3  |
|                                                                               | 10%         | WB     | 37.9  | 36.7  | 38.8  | 43.6  | 50.9  | 57.9  | 62.2  | 63.0  | 61.0  | 55.6  | 50.0  | 42.4  |
|                                                                               |             | MCDB   | 39.6  | 38.2  | 40.8  | 46.3  | 54.0  | 61.9  | 65.7  | 66.6  | 63.2  | 57.3  | 51.8  | 44.2  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 15.8  | 14.9  | 13.1  | 12.5  | 12.6  | 12.9  | 12.1  | 12.9  | 13.2  | 12.8  | 12.5  | 14.0  |
|                                                                               |             | MCDBR  | 19.0  | 14.2  | 14.7  | 17.0  | 18.3  | 19.6  | 17.6  | 17.7  | 15.7  | 13.4  | 12.9  | 16.0  |
|                                                                               |             | MCWBR  | 19.7  | 14.0  | 11.7  | 11.1  | 11.1  | 11.3  | 10.7  | 10.1  | 10.1  | 10.5  | 12.6  | 16.7  |
|                                                                               | 5% WB       | MCDBR  | 19.4  | 14.2  | 12.8  | 13.2  | 15.4  | 16.1  | 15.0  | 13.9  | 12.1  | 10.9  | 12.1  | 16.1  |
|                                                                               |             | MCWBR  | 20.3  | 14.5  | 11.7  | 10.2  | 10.7  | 10.9  | 10.3  | 9.3   | 9.0   | 10.0  | 12.7  | 17.1  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        |        | 0.292 | 0.299 | 0.325 | 0.355 | 0.376 | 0.414 | 0.430 | 0.408 | 0.374 | 0.337 | 0.314 | 0.299 |
|                                                                               | taud        |        | 2.482 | 2.481 | 2.454 | 2.410 | 2.397 | 2.321 | 2.300 | 2.367 | 2.424 | 2.523 | 2.539 | 2.482 |
|                                                                               | Ebn at Noon |        | 264   | 284   | 289   | 287   | 283   | 272   | 265   | 267   | 268   | 265   | 253   | 248   |
|                                                                               | Edh at Noon |        | 22    | 26    | 31    | 35    | 36    | 40    | 40    | 36    | 31    | 25    | 21    | 20    |
| All-Sky Solar<br>Radiation                                                    | RadAvg      |        | 452   | 732   | 1115  | 1392  | 1590  | 1738  | 1769  | 1578  | 1287  | 815   | 474   | 352   |
|                                                                               | RadStd      |        | 42    | 62    | 78    | 131   | 189   | 151   | 142   | 101   | 90    | 82    | 50    | 33    |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | +3.13  | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (29 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | +0.77 | N/A         | N/A   | +97   | N/A   |

Nomenclature: See separate page